



MONASH
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ETC3450 Time Series Econometrics

Assignment 2

Housing Data: Effects of Rezoning Plans

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1 Stationarity

This analysis will be conducted using price data for Melbourne house (house_mel) and apartment (appt_mel) prices, along with Sydney house (house_sydney) and apartment (appt_sydney) prices. These data sets range from Q3 2003 to Q2 2025.

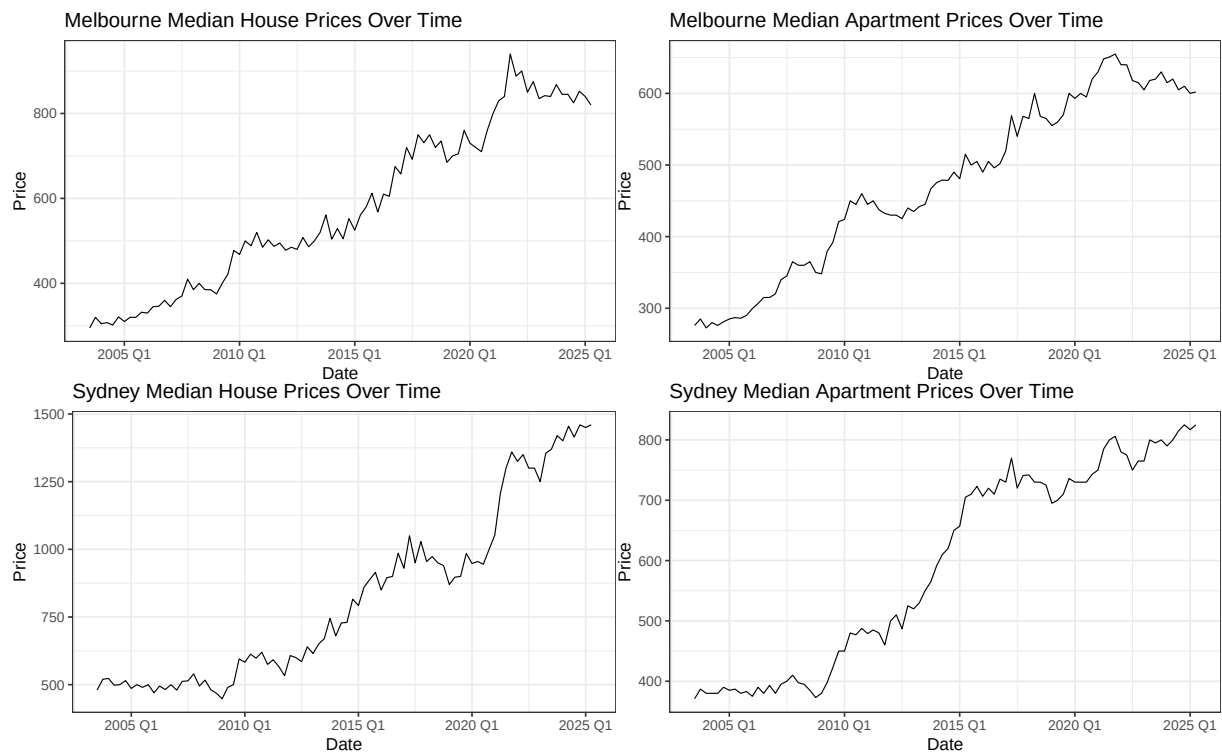
To guide the first steps, it must be determined whether the series mentioned are stationary, or if they have a unit root and require differencing of the series. Only after this is determined can the analysis move onto modelling and forecasting effects.

1.1 Original Series

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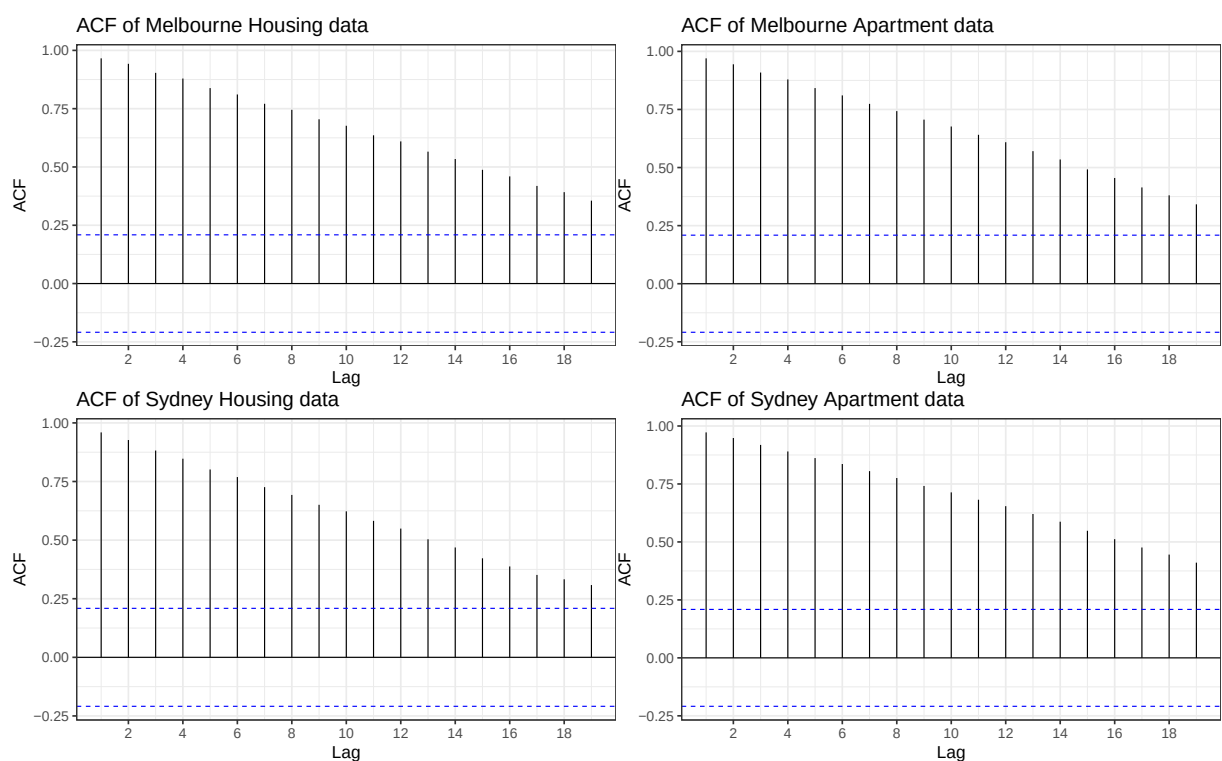
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1.1.1 Time Series Plots

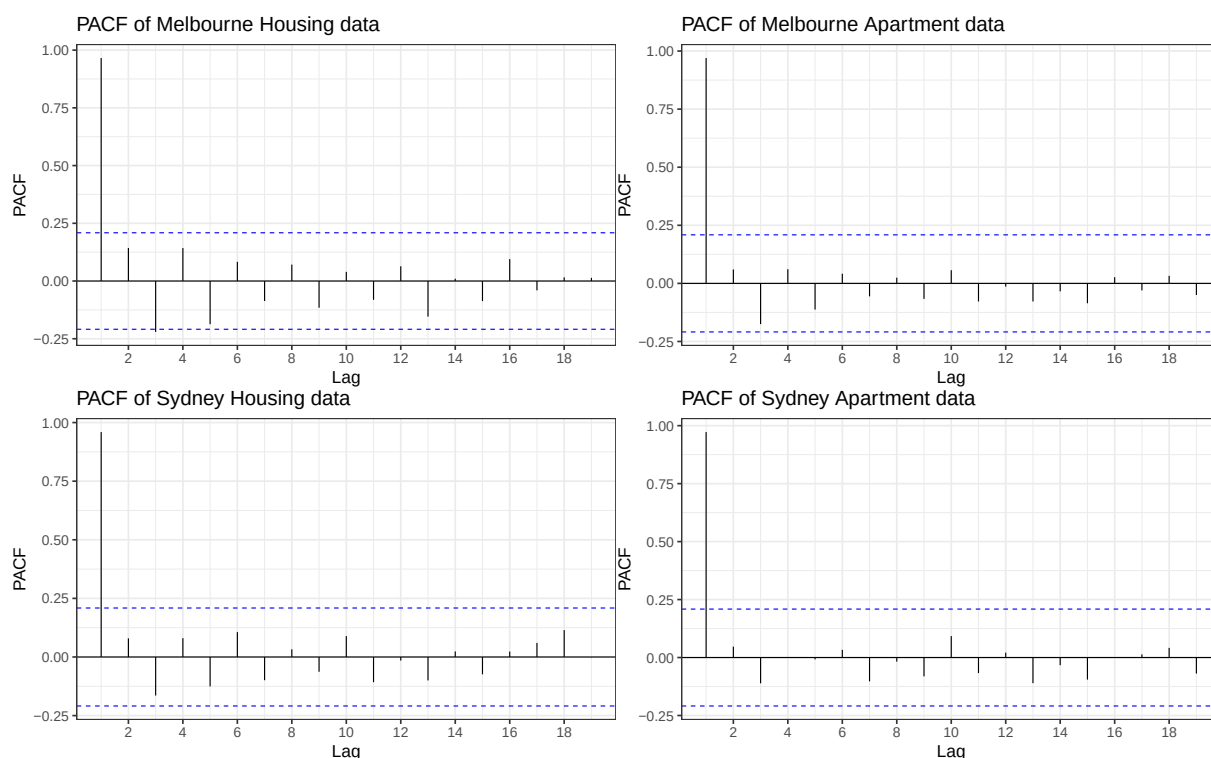


Upon visual inspection of the time series plots in levels, all have an overall positive trend, and seem to be random walks with drift. The means of all series are time-dependent, indicating non-stationary. This is typical of most price data and will guide our reasoning for model specification.

1.1.2 ACF's for Each Series



1.1.3 PACF's for Each Series



The ACF's for all series decay very slowly toward zero, which is characteristic of non-stationary processes such as those with unit roots. This suggests that shocks to the series have persistent effects, and that future values are correlated with past values. This makes intuitive sense for housing prices as they tend to evolve gradually over time rather than fluctuating randomly from period to period, as housing markets exhibit strong inertia and adjustment frictions, such as regulatory and supply constraints.

The PACF's for all series have spikes at the first lag and quickly drop off, suggesting strong dependence on only the observation immediately before. This suggests most of the residual correlation is captured by the first lag which is consistent with a unit root. The spike at the first lag indicates that the series likely becomes stationary after taking first differences, which will remove the unit root. This is typical of housing price data as changes in prices are gradual and are strongly influenced by more recent movements.

1.1.4 Formal Hypothesis tests to determine stationarity (ADF and KPSS):

Table 1: Stationarity Test Results Summary

Metric	House_mel	Appt_mel	House_syd	Appt_syd
ADF Test Stat	-0.6	-1.35	1.07	-0.2

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Metric	House_mel	Appt_mel	House_syd	Appt_syd
ADF Result	Fail to reject	Fail to reject	Fail to reject	Fail to reject
KPSS p-value	0.01	0.01	0.01	0.01
KPSS Result	Reject	Reject	Reject	Reject

The ADF test null is that the series is non-stationary and thus has a unit root. This test is left sided with the null being rejected if the ADF test statistic is less than the critical value. The critical value at the 5% level is -2.89 . As all the test statistics are greater than the critical value, there is insufficient evidence to reject the null, thus conclude insufficient evidence to suggest stationarity in any of the series.

The KPSS test is a reversal of the ADF test, where the null is that the series is stationary, and the alternative is non-stationarity. For this test a p -value is given for its simplicity, where a p -value less than 0.05 results in a rejection. As all p -values are 0.01, the conclusion is that the series are all non-stationary, which supports the conclusion from the ADF.

All series are determined to be non-stationary and have unit roots. According to the PACF's, all are $I(1)$ - first order integrated – meaning that taking the first difference of each series should make each stationary.

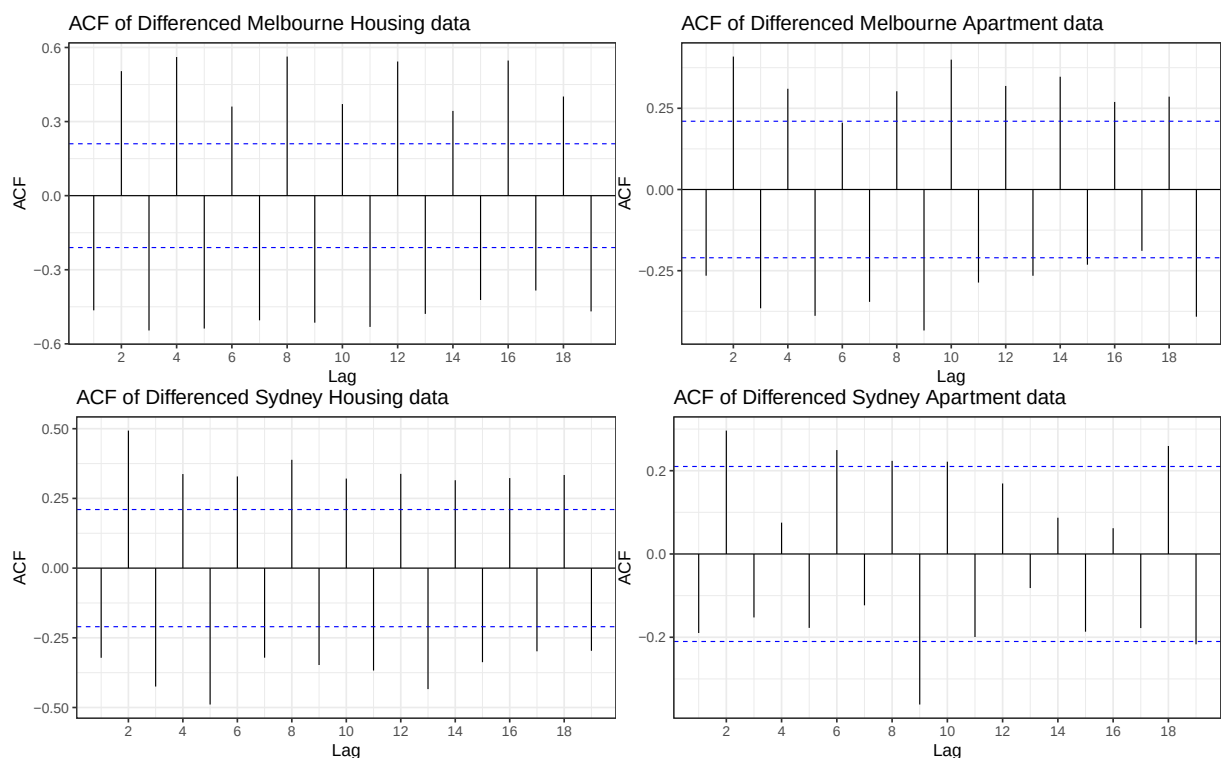
1.2 First Difference Series

1.2.1 Differenced Series Plots

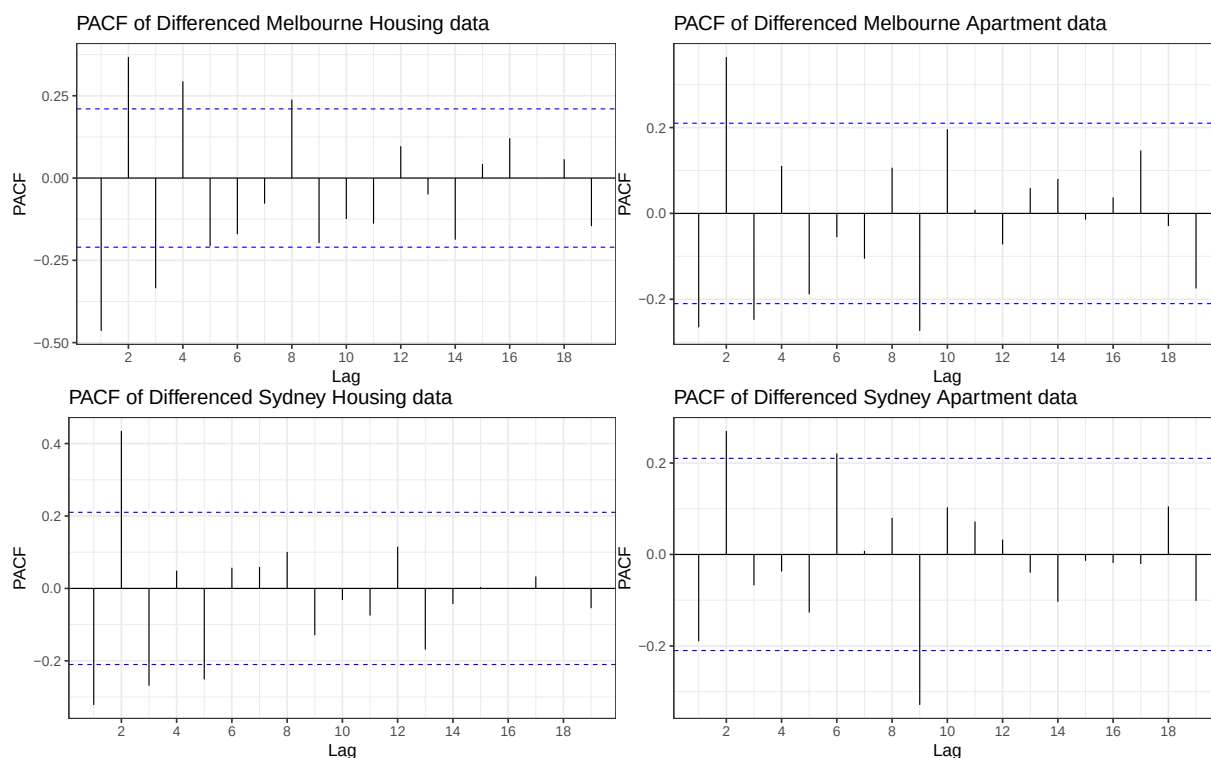


The series of the first order differences looks stationary as there is a constant mean of 0 across time and displays mean reverting behaviour. There are no obvious trends nor seasonality, however there appears to be a period of high volatility from 2015 to roughly 2018 on all plots.

1.3 Differenced series ACF's



1.4 Differenced series PACF's



Looking at the ACF's and PACF's, the series have strange patterns, but this won't be a problem. With the ACF's, there's a big spike at lag 1, then all others hover near 0 with few small spikes spread out. With the PACF's there are small spikes at early lags and then some random fluctuations. This is consistent with stationary processes with short memory; these series likely have an AR(1) component as previously mentioned.

The ACF's show a strong positive first lag correlation, which indicates short run persistence, or momentum, in price changes. This is typical of housing markets and for a model such as a VAR, is captured by adding a lag.

The PACF's display oscillating behaviour. This means that there are small corrections in price changes, for example, price changes this quarter are positively related to the last quarter's changes but then negatively related to the second lag, which is the correction. This is expected of housing markets as when prices start to rise they might rise for a short time, but then the rate of change would slow down likely due to affordability issues.

Overall, the behaviour of the ACF's and PACF's will not cause any issues, provided that the series are all stationary, which we demonstrate using ADF and KPSS tests again.

1.5 Formal Hypothesis tests to determine stationarity (ADF and KPSS):

Table 2: Stationarity Test Results Summary

Metric	House_mel	Appt_mel	House_sydney	Appt_sydney
ADF Test Stat	-5.13	-4.87	-4.61	-5.26
ADF Result	Reject	Reject	Reject	Reject
KPSS p-value	0.1	0.1	0.1	0.1
KPSS Result	Fail to Reject	Fail to Reject	Fail to Reject	Fail to Reject

The results of these tests are the opposite of the earlier results. As a reminder, the ADF null is that the series is non-stationary, and the KPSS null is that the series is stationary.

As the null is rejected in all ADF tests (test statistics are less than the critical value of -2.89), this leads to the conclusion that the series are all stationary. This is further supported by the KPSS nulls not having enough evidence to reject them. Thus, the conclusion is that the differenced series are all stationary and can be modelled. This conclusion also supports the earlier theory that the series are $I(1)$, as the series became stationary after the first difference.

2 Cointegration Relationships

2.1 Formal Tests

As the variables all relate to property markets in the two biggest Australian cities, it might be possible that cointegrating relationships exist.

If two or more variables (e.g. housing prices and apartment prices in Melbourne) are both $I(1)$ as established in the previous section, it is possible that they can still move together to reach a certain long run equilibrium. Economically, this reflects a market tie such as substitute products for houses and apartments that mean these series move together in the long run.

Cointegration implies that though the variables are non-stationary in levels by themselves, there may be a long run equilibrium relationship.

2.1.1 Engle - Granger Test

The Engle-Granger procedure was conducted to test four possible pairs.

This is a test conducted on the long run regressions of each series on their pair.

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For example, let y_t be median house prices in Melb and let x be house prices in Sydney.

Thus, the long run regression is

$$y_t = c + \beta x_t + u_t$$

An ADF test is then run on the residuals u_t of this regression. There is cointegration if the residuals are $I(0)$, i.e. if the ADF null is rejected.

Here are the results:

Table 3: *Engle Granger Cointegration Test*

Relationship	P Value	Result
House prices in Melb and Syd	0.93	Failed to Reject H_0
Apartment prices in Melb and Syd	0.46	Failed to Reject H_0
Apartment prices and House prices in Melb	0.44	Failed to Reject H_0
Apartment prices and House prices in Syd	0.92	Failed to Reject H_0

For all the tested pairs, there was insufficient evidence to reject H_0 . Thus, there is no evidence of cointegration with any pair. The series only move together in the short run and don't move around a common long run equilibrium.

2.1.2 Johansen Test

The Johansen test is used to detect whether a group of $I(1)$ variables are cointegrated, i.e. whether there is one or more long-run equilibrium relationships.

The Vector Error Correction Model that was used will look like the below:

$$\Delta \mathbf{y}_t = \alpha \beta' \mathbf{y}_{t-1} + \Gamma_1 \Delta \mathbf{y}_{t-1} + \mathbf{D}_t + \varepsilon_t$$

Where:

- $\mathbf{y}_t = \begin{bmatrix} \text{house_mel} & \text{house_syd} & \text{appt_mel} & \text{appt_syd} \end{bmatrix}^T$
- $\Delta \mathbf{y}_t = \mathbf{y}_t - \mathbf{y}_{t-1}$ is the differenced series
- $\alpha \in \mathbb{R}^{4 \times r}$ is the “adjustment” matrix (response to disequilibrium)

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- $\beta \in \mathbb{R}^{4 \times r}$ is the “cointegrating vector” matrix (strength of the cointegrating relationships)
 - $D_t \in \mathbb{R}^{4 \times 1}$ includes the constant and seasonal components
 - $\Gamma_1 \in \mathbb{R}^{4 \times 4}$ are the short run dynamics from lagged differences
 - $r = \text{rank}(\alpha\beta')$

The existence of cointegration in these variables depend on matrix “rank” of the $\alpha\beta'$ matrix, which is denoted as Π . In the Johansen test, the rank of Π is given by r , which will determine the number of cointegrating relationships.

This test was conducted as the Maximum Eigenvalue version, using a VAR(2) in levels leading to 1 lag of differences, and including quarterly seasonal dummies.

The reasoning behind using a VAR(2) is that the series are all $I(1)$, so only 1 lag in differences for the VECM is necessary, as a VAR(2) model collapses to a VECM with one lag.

Table 4: *Engle Granger Cointegration Test*

Relationship	Test Statistic	Critical Value (5%)	Result
$r \leq 3$	2.11	9.24	Fail to Reject
$r \leq 2$	5.75	15.67	Fail to Reject
$r \leq 1$	16.09	22.00	Fail to Reject
$r = 0$	26.64	28.14	Fail to Reject

The test fails to be rejected at the first rank, $r = 0$, so it is appropriate to stop and conclude that there are no cointegrating relationships at the 5% level.

This result supports the result from the previous Engle-Granger tests in which no cointegrating pairs were found. While there may be short run relationships, there are no stable long-run equilibrium relationships.

An explanation for no cross-city relationships could be due to city-specific factors such as different state jurisdictions between Melbourne and Sydney creating different property policies and laws, supply and demand differences such as land constraints and migration trends, and separate regional economies. Nation-wide factors may affect properties in both cities, but the differences in regional dynamics mean there is no shared long run equilibrium.

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An explanation for no relationships in same-city properties could be due to different policies regarding apartments and houses, different supply dynamics, eg. apartment supply can rise fast with construction resources being allocated while house construction is much slower due to land availability. Furthermore, houses and apartments may serve different markets, such as houses for families and apartments for investors/young buyers, meaning that macroeconomic shocks may impact each market differently.

2.2 Implications for Modelling

The Johansen test was conducted using maximum eigenvalue statistics. The null of no cointegrating relationships ($r = 0$) could not be rejected at the 5% significance level, suggesting that the four price series do not share any stable long-run equilibrium relationships.

The fact that the variables are $I(1)$ and are not cointegrated will mean a VAR model in first differences will be used to conduct impulse response analysis and discuss rezoning implications. The VAR will only capture short-run changes as no long-run relationship exists.

The VAR(1) model is estimated as

$$\Delta \mathbf{y}_t = \mathbf{c} + \Phi \Delta \mathbf{y}_{t-1} + \varepsilon_t$$

Where

- $\Delta \mathbf{y}_t = \begin{bmatrix} \Delta \text{house_mel} & \Delta \text{house_syd} & \Delta \text{appt_mel} & \Delta \text{appt_syd} \end{bmatrix}^T$
- $\mathbf{c} \in \mathbb{R}^{4 \times 1}$ is the intercept vector
- $\Phi \in \mathbb{R}^{4 \times 4}$ is the coefficients matrix

To further illustrate, the first equation will be:

$$\begin{aligned} \Delta \text{house_mel}_t &= c_1 + \alpha_{1,1} \Delta \text{house_mel}_{t-1} \\ &\quad + \alpha_{1,2} \Delta \text{appt_mel}_{t-1} \\ &\quad + \alpha_{1,3} \Delta \text{house_syd}_{t-1} \\ &\quad + \alpha_{1,4} \Delta \text{appt_syd}_{t-1} \\ &\quad + \varepsilon_{1,t} \end{aligned}$$

Which tells us that quarterly change in Melbourne house prices depends on:

- Its own past change (momentum)
- Past change in Melbourne apartments (substitution effects)
- Past growth in Sydney houses and apartments (cross-city spillovers)
- Random shocks (policy, interest rates)

This model will capture short run relationships and model how changes in each property series are affected by past changes in the others.

2.2.1 What if There Was One or More Cointegrating Relationship

If the Johansen test had different parameters, it is possible that it would conclude that there is one or more cointegrating relationship. Though the test would be mis-specified, it is worth noting the implications for modelling if this was the case.

If the variables are all $I(1)$ but are cointegrated it means:

- There is a long-run equilibrium relationship, as in a linear combination of the variables in stationary $I(0)$
- Even though the series can have short term differences, they move together in the long run and deviations from the equilibrium are gradually corrected

Economically this means that prices in different housing markets move together over time, they don't drift apart forever. As in if Melbourne apartments and houses are cointegrated then over time if houses rise too far above apartment prices they eventually adjust back to a long run ratio.

Regarding the model itself, instead of a VAR(1), a Vector Error Correction VECM model would be used, in this case a VECM(1), with its model form being the same as what was discussed for the Johansen test.

$$\Delta \mathbf{y}_t = \alpha \beta' \mathbf{y}_{t-1} + \Gamma_1 \Delta \mathbf{y}_{t-1} + \mathbf{D}_t + \varepsilon_t$$

The $\alpha \beta'$ or Π matrix is important here as if $\beta' \mathbf{y}_{t-1} > 0$ then house prices are above their long run equilibrium relative to apartments, so the coefficient α_1 will have to correct the market downwards the next period, meaning α_1 will be negative.

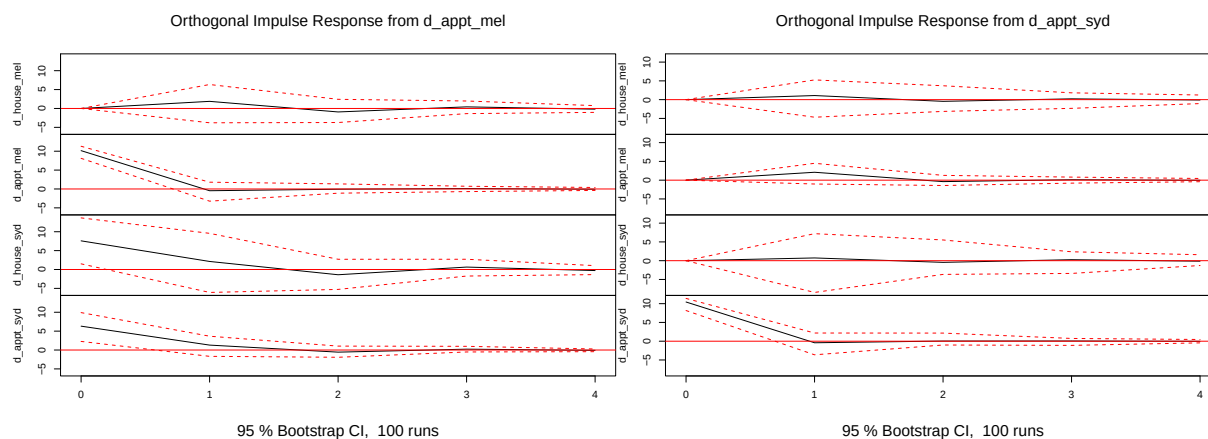
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The long run equilibrium will restore itself through short run dynamics, thus short-run dynamics and long-run equilibrium can be analysed simultaneously.

3 Impulse Response Analysis

3.0.1 Impulse Response Plots for VAR(1)



We found, through a series of tests, that all four series are $I(1)$ and that there are no cointegrating relationships. Thus, a VAR on the first differences of the four series was modelled. The variables in the VAR therefore represent quarterly changes in prices; and the IRFs show how growth rates respond to shocks (short-run effects).

A VAR(1) model was used to compute IRFs with 95% bootstrap CIs and 100 replications, for shocks to apartment prices in both cities.

The ordering of the variables was opposite city houses, opposite city apartments, local city houses, local city apartments. However, it was determined that the ordering did not matter as the same conclusions were reached with other orders.

AIC tests generally suggested lags above 10 were better, however we have relatively few observations (81) and a high number of variables (4), meaning this would create a highly complex model. Thus, VAR(1) was chosen for its parsimony and simplicity.

The sample horizon was chosen as 4 quarters as it was found that the effects of shocks only took one quarter to correct. Thus, to improve readability a period of 1 year was adequate.

Impulse responses in the VAR were based on orthogonalised shocks. They show how each variable in the VAR reacts over time to an unexpected shock in one of the other variables. These shocks are treated as separate, one-off disturbances to each market. For example, an unexpected rise in

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Melbourne apartment prices, while holding other shocks constant. The responses show how property price growth adjusts following that disturbance.

3.0.2 Key Findings

Regarding the impulse response functions, it is important to note that the effect will be the opposite sign, as the policy will create a negative shock while the IRF's only show the effect of a positive shock.

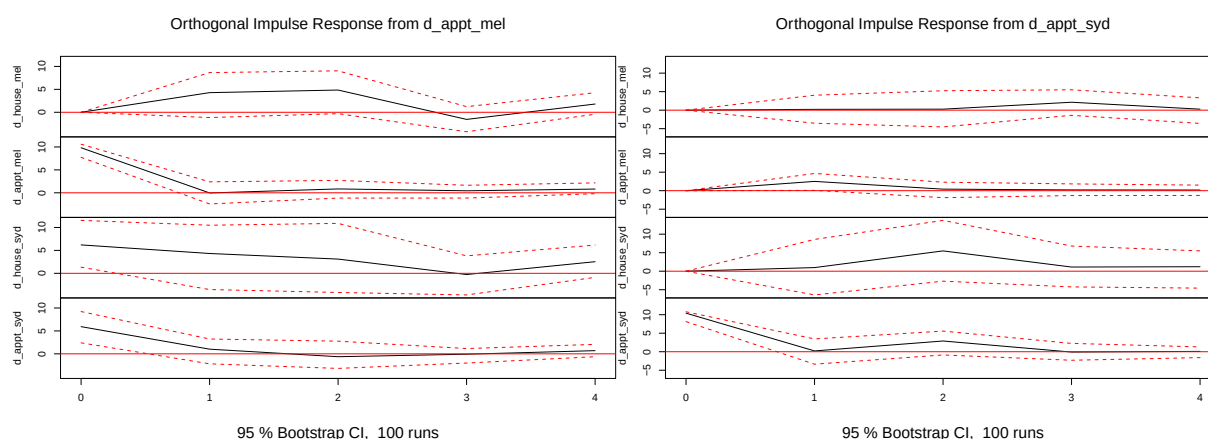
A one standard deviation negative shock to Melbourne apartment price changes produces a negative immediate response in Melbourne apartment prices. This effect dissipates after one quarter. For Melbourne house prices the reaction is statistically insignificant, likely due to buyers not immediately switching between properties due to a shock in one. Reasons likely being costs, locations, and target demographics. For Sydney markets the response is also insignificant, likely due to differences and distance between states.

A one standard deviation negative shock to Sydney apartment price changes produces no response in Sydney houses, likely due to the previously mentioned reasons about buyers not immediately switching from buying houses to apartments. The biggest response is to its own market, as this shock causes an immediate fall in Sydney apartment prices which then dissipates after a quarter.

Overall, the spillovers across cities and housing types are small and short-lived, the stronger responses are for the own variable, which is consistent with findings of no long-run cointegrating relationships.

3.1 What if the Number of Lags was Changed?

3.1.1 Impulse Response Plots for VAR(2)



Moving from VAR(1) to VAR(2) didn't significantly alter the impulse response functions. The biggest shock was still to the same variable, i.e. a shock to Melbourne apartments will have the biggest effect on its own series. The shocks to other series were still insignificant.

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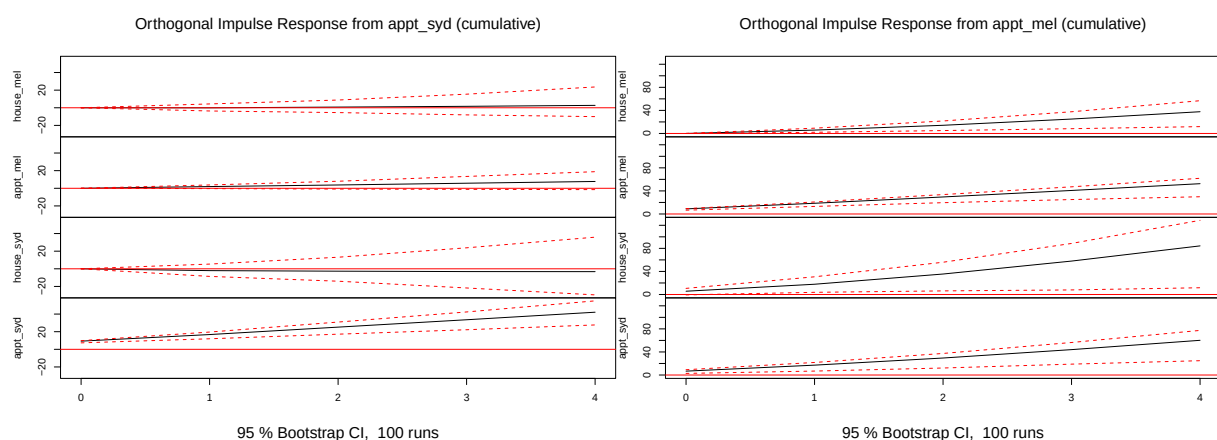
This suggests that most of the dynamics were captured within one lag. The housing markets respond mainly to recent changes and including an extra lag doesn't add new information.

Once the series are differenced to achieve stationarity, price changes don't strongly depend on changes two quarters ago.

The system seems stable, and the VAR(1) is parsimonious and sufficient for the sample size. This is further demonstrated by the ACF's of the residuals of the VAR(2), which has a spike at lag 1 then quickly declines to 0, indicating little persistence beyond one quarter and not much serial correlation.

3.2 What if we Used a Different Model?

3.2.1 Impulse Response for VECM(1)



Changing the number of lags in the Johansen test from 2 to 3 gave a result of one cointegrating relationship at the 5% level. Though this would usually mean sufficient evidence, the facts that the test barely passes at this level and that the Engle-Granger tests suggested no cointegrating relationships, this is not the main model for analysis. If there is a cointegrating relationship, then a VECM is much more appropriate than a VAR model as it can account for a long-run equilibrium.

Cumulative IRF's can display long run relationships and equilibrium better than non-cumulative IRF's, which is why they are used here.

From the cumulative IRF's, once again the most significant change is to the variable's own series. A shock to Sydney apartments leads to a gradual but persistent rise in Sydney apartment prices, with this being the same for a Melbourne apartment shock. The fact that the cumulative lines don't flatten out means that the long-run equilibrium relationship is weak or slow. The model behaves like a VAR, which is the model the further analysis is based on.

Effects are insignificant on other markets, which matches the earlier findings. These IRF's reinforce that the short-run dynamics are limited and local.

4 Short-Term and Long-Term Effects of the Policy

4.1 Short-Term (< 1 Quarter)

The rezoning policy is aimed at increasing the supply of apartments, thus creating a negative shock to apartment prices in Melbourne and Sydney.

This policy will cause an immediate decline in apartment prices; however, this is very transitory. The effect will dissipate within a quarter. The policy changes growth rates briefly but the effect fades quickly.

There are also negligible spillovers. There is no statistically significant immediate effect on local house prices or opposite-city property markets. Short run price falls in apartments do not translate into cross-market or cross-city transmissions in the quarters after the shock.

If the policy's goal is a short, fast fall in prices then it may work, but only briefly and only in the targeted market.

4.2 Medium to Long-Term Effects (> 1 Quarter)

As the Johansen and Engle Granger tests found no cointegration, and the IRF's dissipate quickly, the shock does not produce a lasting change in the relative levels of prices across markets.

The absence of cointegration implies the markets do not revert to a new joint equilibrium following the shock, local levels return to their prior paths rather than settling at a new equilibrium.

If the objective is a sustained or structural change in the housing system, eg. permanently higher affordability, then a one off shock is insignificant. Persistent interventions such as tax reforms or land-use policy reforms would be required.

These conclusions are the same for all 3 models used.

5 Limitations

An issue with this analysis is the small sample size. The data covers quarterly prices from 2003 to 2025, which has fewer statistical power than higher frequency data such as monthly data. Monthly data will also allow a monthly analysis of IRF's to see exactly how long shocks persist rather than quarter-by-quarter predictions.